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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays00:27

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

```
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Array2:")
8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (hstack)
11 # Perform horizontal stacking (side by side)
```

Average time0.030 s30.50 ms

Maximum time0.042 s42.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 142 ms

Debug

Expected output

Enter Array1:

1 2 3

4 5 6

7 8 9

Enter Array2:

4 5 6

Actual output

Enter Array1:

1 2 3

4 5 6

7 8 9

Enter Array2:

4 5 6

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Test cases

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3.2.1. Numpy: Matrix Operations

06:31

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The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. Addition (`matrix_a + matrix_b`)

2. Subtraction (`matrix_a - matrix_b`)

3. Element-wise Multiplication (`matrix_a * matrix_b`)

4. Matrix Multiplication (`matrix_a · matrix_b`)

5. Transpose of Matrix A

Input Format:

The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.

Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.

2. The result of Subtraction.

3. The result of Element-wise Multiplication.

4. The result of Matrix Multiplication.

5. The Transpose of Matrix A.

matrixOp...

1

import numpy as np

2

3

Input matrices

4

print("Enter Matrix A:")

5

matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

6

7

print("Enter Matrix B:")

8

matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

9

Average time

0.024 s

23.60 ms

Maximum time

0.034 s

34.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

34 ms

Debug

Expected output

Enter Matrix A:

1 2 3

4 5 6

7 8 9

Enter Matrix B:

1 1 1

Actual output

Enter Matrix A:

1 2 3

4 5 6

7 8 9

Enter Matrix B:

1 1 1

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Test cases

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3.1.1.1. Numpy Array Operations

00:38

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases

numpyarr...

1 import numpy as np

2

3 # write your code here...

4 rows, cols = map(int,input().split())

5 if rows>0 and cols>0:

6 elements = []

7 for i in range(rows):

8 elements.extend(map(int,input().split()))

9 arr = np.array(elements).reshape(rows,cols)

10 else : arr = np.array([]).reshape(rows,cols)

11 print(arr)

Average time 0.013 s 13.50 ms

Maximum time 0.023 s 23.00 ms

2 out of 2 shown test case(s) passed

10 out of 10 hidden test case(s) passed

Test case 1 22 ms Debug

Expected output

3 4

1 2 3 4

5 6 7 8

9 10 11 12

[[1 2 3 4]

[5 6 7 8]

Actual output

3 4

1 2 3 4

5 6 7 8

9 10 11 12

[[1 2 3 4]

[5 6 7 8]

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3.2.7. Student Data Analysis and Operations

01:52

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.

Find total students: Determine the total number of students in the dataset.

Print all student roll numbers: Extract and print the roll numbers of all students.

Print Subject 1 marks: Extract and print the marks of all students in Subject 1.

Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.

Find maximum marks in Subject 3: Identify the highest marks in Subject 3.

Print all subject marks: Display the marks of all students for each subject.

Find total marks of students: Compute the total marks for each student across all subjects.

Find the average marks of each student: Compute the average marks for each student.

Find average marks of each subject: Compute the average marks for all students in each subject.

Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.

Find average marks of Subject 1 and Subject 3: Compute the average marks for Subject 1 and Subject 3.

Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.

Find the roll number of the student with minimum marks in Subject 2: Identify the student with the lowest marks in Subject 2 and print their roll number.

Find the roll number of students who scored 24 marks in Subject 2: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.

Find the count of students who got less than 40 marks in Subject 1: Count the number of

Operation...

1import numpy as np

2

3a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

4

5# 1. Print all student

6# 1. Print all student details

7print("All student Details:\n",a)

8

9# 2. print total students

10

11print("Total Students:", len(a))

Average time

0.018 s

18.00 ms

Maximum time

0.018 s

18.00 ms

1 out of 1 shown test case(s) passed

Test case 1

18 ms

Debug

Expected output

All student Details:

[[301. 67. 77. 88.]

[302. 78. 88. 77.]

[303. 45. 56. 89.]

[304. 88. 98. 45.]

[305. 78. 88. 99.]

Actual output

All student Details:

[[301. 67. 77. 88.]

[302. 78. 88. 77.]

[303. 45. 56. 89.]

[304. 88. 98. 45.]

[305. 78. 88. 99.]

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Test cases

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3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

00:32

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- Searching:** Find the indices where search_value appears in array1 and print these indices.
- Counting:** Count how many times count_value appears in array1 and print the count.
- Broadcasting:** Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
- Sorting:** Sort array1 in ascending order and print the sorted array.

Input Format:

- A single line containing space-separated integers representing array1.
- An integer search_value represents the value to search for in the array.
- An integer count_value represents the value to count in the array.
- An integer broadcast_value represents the value to add to each element of the array.

Output Format:

- The indices where search_value occurs in array1.
- The count of occurrences of count_value in array1.
- The array after adding the broadcast_value to each element.

arrayOpe...

Submit

1 import numpy as np

2

3 # Input array from the user

4 array1 = np.array(list(map(int, input().split())))

5

6 # Searching

7 search_value = int(input("Value to search: "))

8 count_value = int(input("Value to count: "))

9 broadcast_value = int(input("Value to add: "))

10

11 # Find indices where value matches in array1

Average time 0.015 s 15.25 ms

Maximum time 0.022 s 22.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 22 ms Debug

Expected output

1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0 1 2]

3

Actual output

1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0 1 2]

3

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3.2.5. Numpy: Copying and Viewing Arrays

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

Creating a view of the original_array and assigning it to view_array.

Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

A single line of space-separated integers.

Output Format:

After modifying the view:

Original array after modifying view: <original_array>
View array: <view_array>

After modifying the copy:

Original array after modifying copy: <original_array>
Copy array: <copy_array>

Sample Test Cases

copyAnd...

```
1 import numpy as np
2
3 inputlist = list(map(int,input().split(" ")))
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array =original_array.view()
10
11 # Create a copy
```

Average time

0.011 s

Maximum time

0.021 s

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

10 20 30 40 50 60 70 80

Actual output

10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]

Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operations

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

• Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

• Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

• Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

• The first line contains space-separated integers representing the elements of array A.

• The second line contains space-separated integers representing the elements of array B.

Output Format:

• For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

```
1 import numpy as np
2
3 def array_operations(A, B):
4
5     # Convert A and B to NumPy arrays
6     arr_A = np.array(A)
7     arr_B = np.array(B)
8
9     # Arithmetic Operations
10    sum_result = arr_A + arr_B # Element-wise sum
11    diff_result = arr_A - arr_B # Element-wise difference
```

Average time

0.011 s

10.67 ms

Maximum time

0.015 s

15.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

15 ms

Debug

Expected output

1 2 3 4

5 6 7 8

Element-wise Sum: 6 8 10 12

Element-wise Difference: -4 -4 -4 -4

Element-wise Product: 5 12 21 32

Mean of A: 2.5

Actual output

1 2 3 4

5 6 7 8

Element-wise Sum: 6 8 10 12

Element-wise Difference: -4 -4 -4 -4

Element-wise Product: 5 12 21 32

Mean of A: 2.5

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Test cases

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3.2.3. Numpy: Custom Sequence Generation

00:31

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customS...

1 import numpy as np

2

3 # Take user input for the start, stop, and step of the sequence

4 start = int(input())

5 stop = int(input())

6 step = int(input())

7

8 # Generate the sequence using np.arange()

9

10 # Print the generated sequence

11 # Generate the sequence using np.arange()

Average time 0.012 s 12.25 ms

Maximum time 0.015 s 15.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 15 ms Debug

Expected output

3

10

2

[3 5 7 9]

Actual output

3

10

2

[3 5 7 9]

Test case 2 4 ms

Terminal Test cases

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